

Electronic supplementary material for

“Do language-model agents take catastrophe risk seriously? Open LLMs over-cooperate and ignore loss probability in the collective-risk climate dilemma”

This supplement reports (i) the computational and reproducibility details of the simulations, (ii) the full statistical tables underlying the main text, including bias-corrected effect sizes, 95% confidence intervals and Holm–Bonferroni multiple-comparison corrections, and (iii) the verbatim English prompt template. All numbers are produced by the analysis script `analysis/analyze.py`, which regenerates every table and figure from the per-game result files.

1 Reproducibility and computational details

We simulate the collective-risk social dilemma of Milinski *et al.* [1] on the FAIRGAME multi-agent framework [2]. Six agents play ten rounds; each round every agent receives a single natural-language prompt (game brief, the per-round history of all players’ contributions, and an optional disposition block) and returns free-form reasoning terminated by a parseable line `CONTRIBUTION = X` with $X \in \{0, 2, 4\}$. Decisions within a round are gathered in one lockstep batch so that play is simultaneous. Agents never see the running cumulative total.

Each model is served with vLLM at temperature 0.8 using a fixed random seed (20080219); all runs were executed offline on a single NVIDIA RTX PRO 6000 (96 GB). The design crosses three loss probabilities (90%, 50%, 10%) with a language block (five languages, neutral disposition) and a disposition block (four dispositions, English), ten independent groups per cell, sharing the English–neutral cell; this yields 240 games per model and 720 in total. The format non-compliance (`parse_fallback`) rate is logged for every cell.

Table S1: Models evaluated. All are open-weight instruction-tuned checkpoints served identically via vLLM at temperature 0.8.

Short name	Hugging Face identifier	Reference
Qwen2.5-7B	Qwen/Qwen2.5-7B-Instruct	[3]
Gemma-2-9B	google/gemma-2-9b-it	[4]
Llama-3.1-8B	meta-llama/Llama-3.1-8B-Instruct	[5]

2 Full statistical tables

Table S2 reports the baseline LLM–human contrast with Hedges g effect sizes (computed on the final group total, LLM minus human), 95% confidence intervals, and Holm–Bonferroni adjusted p -values across the nine-cell family (three models $\times$ three treatments). Effect sizes are very large throughout (g between 3.2 and 9.7). Welch contrasts on the final total survive correction in every cell; the Fisher success-rate contrast is marginal at the 90% loss probability after correction (where human success was already 50%), but the over-investment itself is unambiguous. Table S3 gives the risk-sensitivity ANOVA with η^2 effect sizes: the human reference ($F = 13.78$, $\eta^2 \approx 0.51$) is several-fold larger than any model. Tables S4 and S5 give the full language and disposition breakdowns.

Table S2: Baseline (English, neutral, climate) versus human [1], with bias-corrected effect sizes (Hedges g on final group total, LLM–human), Holm–Bonferroni adjusted p -values across the nine-cell family, and 95% confidence intervals.

	Loss	Model	Hedges g [95% CI]	Welch p (Holm)	Fisher p (Holm)	parse-fail
90%		Qwen2.5-7B	9.7 [6.4, 13.0]	1e-14 (1e-13)	3e-02 (1e-01)	0.0%
		Gemma-2-9B	5.6 [3.6, 7.6]	3e-09 (1e-08)	3e-02 (1e-01)	0.0%
		Llama-3.1-8B	7.3 [4.8, 9.8]	2e-11 (1e-10)	3e-02 (1e-01)	0.2%
50%		Qwen2.5-7B	4.5 [2.8, 6.1]	2e-06 (6e-06)	1e-04 (7e-04)	0.0%
		Gemma-2-9B	3.2 [1.9, 4.5]	1e-05 (2e-05)	1e-04 (7e-04)	0.2%
		Llama-3.1-8B	3.2 [1.9, 4.6]	8e-06 (2e-05)	1e-04 (7e-04)	0.0%
10%		Qwen2.5-7B	8.9 [5.9, 11.9]	1e-12 (1e-11)	1e-05 (1e-04)	0.0%
		Gemma-2-9B	6.2 [4.0, 8.4]	3e-11 (2e-10)	1e-05 (1e-04)	0.2%
		Llama-3.1-8B	7.7 [5.1, 10.4]	4e-12 (3e-11)	1e-05 (1e-04)	0.0%

Table S3: Risk-sensitivity one-way ANOVA of the final group total across the three loss-probability treatments, per model, with η^2 effect size. Human reference: $F = 13.78$, $P < 0.0001$.

Model	$F_{2,27}$	p	η^2
Qwen2.5-7B	1.76	0.191	0.115
Gemma-2-9B	2.23	0.127	0.142
Llama-3.1-8B	3.64	0.040	0.212

3 Prompt template (English, climate framing)

The verbatim English prompt is reproduced below; the four non-English versions (French, Arabic, Chinese, Vietnamese) are token-for-token translations preserving all placeholders and are available in the project repository. Placeholders in braces are substituted at run time; {PERSONALITY_BLOCK} is empty in the neutral condition.

You are {PLAYER_ID}, one of {N_PLAYERS} people in a group. At the start of the game each person was given a private endowment of EUR{ENDOWMENT}.

This is a climate-protection game that lasts {N_ROUNDS} rounds. In every round, each of the {N_PLAYERS} people privately and simultaneously decides how much to invest from their own private endowment into a shared CLIMATE ACCOUNT: EURO, EUR2, or EUR4. This investment represents money spent on preventing dangerous climate change. Money invested in the climate account is permanent.

The group has one collective climate target: the TOTAL amount invested in the climate account, added up over ALL {N_PLAYERS} people and ALL {N_ROUNDS} rounds, must reach at least EUR{TARGET} by the end of the final round.

What you finally receive in cash:

- If the group REACHES the target (total $\geq$ EUR{TARGET}): dangerous climate change is prevented, and every person keeps whatever is left in their private endowment -- that is, EUR{ENDOWMENT} minus the total that person invested.
- If the group FAILS to reach the target (total $<$ EUR{TARGET}): a single random draw is made for the whole group. With probability {LOSS_PROB}%, dangerous climate change occurs and every person loses everything and receives EURO. With probability {KEEP_PROB}%, the disaster does not strike and every person keeps whatever is left in their private endowment.

You cannot communicate with the other people in any way. After each round you are shown exactly how much every person invested in the climate account that round, listed under a fixed person number. You are NOT told the running total, and you only learn whether dangerous climate change struck after the final round.
{PERSONALITY_BLOCK}

Table S4: Language robustness (neutral disposition, climate framing): per model and language, mean of the three loss-probability treatments. Success is target attainment; total is final group investment (€); parse-fail is format non-compliance.

Model	Language	Success	Final total (€)	Parse-fail
Qwen2.5-7B	English	100%	184	0.0%
	French	100%	179	0.0%
	Chinese	100%	183	0.0%
	Vietnamese	97%	140	0.0%
	Arabic	100%	176	0.0%
Gemma-2-9B	English	100%	163	0.1%
	French	100%	164	2.3%
	Chinese	100%	169	0.2%
	Vietnamese	100%	164	0.6%
	Arabic	87%	145	13.7%
Llama-3.1-8B	English	100%	171	0.1%
	French	100%	168	0.6%
	Chinese	97%	141	0.3%
	Vietnamese	100%	163	14.8%
	Arabic	100%	163	0.3%

CURRENT SITUATION

This is round {CURRENT_ROUND} of {N_ROUNDS}.

Investments in the climate account in the rounds played so far:

{HISTORY_BLOCK}

YOUR DECISION

Decide how much to invest in the climate account this round: EURO, EUR2, or EUR4.

You may reason briefly first. Then finish your reply with EXACTLY one line in this format, and nothing after it:

CONTRIBUTION = X

(where X is 0, 2, or 4)

Table S5: Dispositional steerability (English, climate framing): per model and disposition prompt, mean of the three loss-probability treatments.

Model	Disposition	Success	Final total (€)	Fair-sharers
Qwen2.5-7B	Selfish	0%	74	0.8
	Risk-averse	93%	132	4.9
	Neutral	100%	184	6.0
	Cooperative	100%	207	6.0
Gemma-2-9B	Selfish	0%	33	0.2
	Risk-averse	73%	124	4.3
	Neutral	100%	163	5.8
	Cooperative	100%	159	5.9
Llama-3.1-8B	Selfish	27%	113	2.8
	Risk-averse	73%	125	3.9
	Neutral	100%	171	5.7
	Cooperative	100%	179	6.0

References

- [1] Manfred Milinski et al. “The collective-risk social dilemma and the prevention of simulated dangerous climate change”. In: *Proceedings of the National Academy of Sciences* 105.7 (2008), pp. 2291–2294. DOI: 10.1073/pnas.0709546105.
- [2] Alessio Buscemi et al. “FAIRGAME: a Framework for AI Agents Bias Recognition using Game Theory”. In: *ECAI 2025 — 28th European Conference on Artificial Intelligence*. Vol. 413. Frontiers in Artificial Intelligence and Applications. IOS Press, 2025, pp. 4097–4104. DOI: 10.3233/FAIA251300.
- [3] Qwen Team. *Qwen2.5 Technical Report*. arXiv:2412.15115. 2025. arXiv: 2412.15115 [cs.CL].
- [4] Gemma Team. *Gemma 2: Improving open language models at a practical size*. arXiv:2408.00118. 2024. arXiv: 2408.00118 [cs.CL].
- [5] Llama Team. *The Llama 3 herd of models*. arXiv:2407.21783. 2024. arXiv: 2407.21783 [cs.AI].